1. The 2’s complement of the 8-bit binary `11001010` is:
   a) `00110101`
   b) `00110110`
   c) `11001011`
   d) `00110100`
   Answer: b
2. The range of decimal integers that can be represented in 8-bit 2’s complement is:
   a) –127 to +127
   b) –128 to +127
   c) –127 to +128
   d) –256 to +255
   Answer: b
3. In IEEE 754 single-precision format, the mantissa (fraction) uses how many bits?
   a) 22
   b) 23
   c) 24
   d) 31
   Answer: b
4. In IEEE 754 single-precision, the exponent is stored in:
   a) Sign-magnitude
   b) 1’s complement
   c) 2’s complement
   d) Biased (excess-127) form
   Answer: d
5. The RTL statement `R1 ← R2 + R3` means:
   a) Transfer R2 to R1 and add R3
   b) Add contents of R2 and R3 and store result in R1
   c) Move R1 to R2 and R3
   d) XOR R2 and R3 into R1
   Answer: b
6. Which of the following is a *logic* micro-operation?
   a) `R1 ← R2 + R3`
   b) `R1 ← R2 OR R3`
   c) `R1 ← R2 - 1`
   d) `R1 ← R2 ← shift left`
   Answer: b
7. Shift micro-operations are mainly used for:
   a) Arithmetic and logical operations, addressing
   b) Only data storage
   c) Only instruction decoding
   d) Only I/O operations
   Answer: a
8. A shared bus with tri-state buffers helps reduce:
   a) Clock frequency
   b) Number of physical wires
   c) Power supply capacity

d) Memory size

Answer: b

9. In RTL, the statement `R1 ← R2 ASHL` represents:
   a) Arithmetic shift right of R2 into R1
   b) Arithmetic shift left of R2 into R1
   c) Logical shift left of R2 into R1
   d) Logical shift right of R2 into R1

   Answer: b

10. Which of the following is *not* a basic data type in COA?
    a) Integer
    b) Floating-point
    c) Boolean
    d) Method

    Answer: d

11. Changing a number from one radix to another is called:
    a) Encoding
    b) Number conversion
    c) Truncation
    d) Rounding

    Answer: b

12. The 2’s complement representation is preferred because it:
    a) Makes unsigned arithmetic easier
    b) Has only one representation of zero and simplifies subtraction
    c) Reduces the number of bits
    d) Avoids overflow

    Answer: b

13. Overflow in 2’s complement addition is detected by:
    a) Sign of the result only
    b) Carry-in and carry-out of the sign bit
    c) Comparing operands and result
    d) Using parity

    Answer: b

14. The BCD (Binary Coded Decimal) representation of decimal `9` in 4 bits is:
    a) `0101`
    b) `1001`
    c) `1010`
    d) `1111`

    Answer: b

15. In RTL, `R1 ← R2 ∧ R3` denotes:
    a) OR
    b) AND
    c) XOR
    d) NOT

    Answer: b

16. The main purpose of using a bus with tri-state buffers is:
    a) To allow only one device to drive the bus at a time
    b) To increase bus speed
    c) To reduce clock skew

d) To increase memory size
Answer: a

17. A control bus normally carries:
    a) Data
    b) Memory addresses
    c) Control signals
    d) Power
    Answer: c
18. Which of the following is *not* a micro-operation class?
    a) Arithmetic
    b) Logic
    c) Shift
    d) Encryption
    Answer: d
19. The sign bit in 2’s complement format is located at:
    a) LSB
    b) Second bit
    c) MSB
    d) Any bit
    Answer: c
20. Binary representation of decimal `15` in 4 bits is:
    a) `1010`
    b) `1100`
    c) `1111`
    d) `1001`
    Answer: c
21. In IEEE 754, the sign bit is:
    a) Bit 0
    b) Bit 1
    c) Bit 31
    d) Bit 32
    Answer: a
22. The decimal equivalent of hexadecimal `A3` is:
    a) 153
    b) 163
    c) 173
    d) 183
    Answer: b
23. The hexadecimal equivalent of binary `11010110` is:
    a) `D6`
    b) `D7`
    c) `E6`
    d) `E7`
    Answer: a
24. A 16-bit address bus can address up to:
    a) 32 KB
    b) 64 KB
    c) 128 KB

d) 256 KB
Answer: b
25. The size of a 32-bit data bus is:
a) 4 bytes
b) 8 bytes
c) 16 bytes
d) 32 bytes
Answer: a
26. Which of the following is a *logical* shift operation?
a) `LSL` (Logical Shift Left)
b) `ASL` (Arithmetic Shift Left)
c) `ASR` (Arithmetic Shift Right)
d) `ROR` (Rotate Right)
Answer: a
27. The RTL notation `R1 ← R2′` means:
a) Complement of R2 into R1
b) Increment R2 by 1
c) Decrement R2 by 1
d) Load R1 from R2
Answer: a
28. Gray code is mainly used in:
a) Decimal arithmetic
b) Floating-point arithmetic
c) Position encoders and error minimization
d) Cache mapping
Answer: c
29. The number of bits in the IEEE 754 single-precision format is:
a) 16
b) 32
c) 64
d) 128
Answer: b
30. The 2's complement of `00000001` is:
a) `11111110`
b) `11111111`
c) `00000000`
d) `10000000`
Answer: b

---

31. Which register holds the address of the next instruction to be fetched?
a) IR
b) MDR
c) PC
d) MAR
Answer: c
32. The sequence "Fetch, Decode, Execute" is called:
a) DMA cycle
b) Interrupt cycle
c) Instruction cycle

d) Memory cycle
Answer: c

33. The IR (Instruction Register) stores:
a) Address of the current memory word
b) Current instruction being executed
c) ALU result
d) Status bits
Answer: b

34. The MAR (Memory Address Register) holds:
a) Data to be written
b) Address of the memory location
c) Instruction opcode
d) ALU result
Answer: b

35. The MDR (Memory Data Register) holds:
a) Memory address
b) Data being read from or written to memory
c) Program counter value
d) Status flags
Answer: b

36. In a typical computer, the address bus carries:
a) Data
b) Memory addresses
c) Control signals
d) Interrupt vectors only
Answer: b

37. The data bus is:
a) Unidirectional
b) Bidirectional
c) Only input
d) Only output
Answer: b

38. The control unit in the basic computer design is responsible for:
a) Only ALU operations
b) Generating timing and control signals
c) Only memory refresh
d) Only I/O mapping
Answer: b

39. The ALU mainly performs:
a) Data storage
b) Arithmetic and logic operations
c) Instruction fetching
d) I/O control
Answer: b

40. The main memory unit is also called:
a) Cache
b) Main store / RAM
c) Register file
d) Control memory
Answer: b

41. The CPU is composed of:
    a) CPU, memory, I/O
    b) ALU, registers, control unit
    c) Only ALU
    d) Only memory
    Answer: b
42. Which of the following is *not* a CPU register?
    a) PC
    b) IR
    c) MAR
    d) GPU
    Answer: d
43. Time required for one instruction cycle is known as:
    a) Clock period
    b) Machine cycle
    c) Instruction time
    d) Memory cycle
    Answer: c
44. The bus that carries read/write signals is:
    a) Address bus
    b) Data bus
    c) Control bus
    d) Power bus
    Answer: c
45. A memory-mapped I/O system treats I/O devices as:
    a) Separate from memory
    b) Part of the memory address space
    c) Only controlled by cache
    d) Directly connected to ALU
    Answer: b
46. Which of the following is true for a von Neumann architecture?
    a) Separate buses for data and instructions
    b) Shared memory for data and instructions
    c) Only stack-based execution
    d) Only Harvard-style CPU
    Answer: b
47. The term “bus width” usually refers to the number of:
    a) Control lines
    b) Address lines
    c) Data lines
    d) Power lines
    Answer: c
48. The size of the data bus affects the:
    a) Number of addressable memory locations
    b) Width of data transferred per cycle
    c) Clock speed
    d) Instruction length
    Answer: b
49. The size of the address bus affects the:
    a) Bus clock frequency

b) Maximum addressable memory size
c) Data transfer rate
d) Number of registers
Answer: b

50. Interrupts are mainly used to:
a) Increase CPU speed
b) Allow CPU to respond to external events
c) Replace memory
d) Replace ALU
Answer: b

51. DMA (Direct Memory Access) is used to:
a) Increase instruction complexity
b) Allow I/O devices to transfer data directly to memory without CPU
c) Only store microprograms
d) Replace control unit
Answer: b

52. Which of the following is *not* a basic functional unit?
a) CPU
b) Memory
c) I/O
d) ALU only
Answer: d

53. The accumulator is a type of:
a) I/O port
b) Special-purpose CPU register
c) Disk sector
d) Cache line
Answer: b

54. A typical “instruction cycle” has:
a) Only fetch phase
b) Only execute phase
c) Fetch, decode, execute, and sometimes interrupt
d) Only interrupt phase
Answer: c

55. The term “bus arbitration” refers to:
a) Assigning priorities to different bus masters
b) Increasing bus speed
c) Decreasing data width
d) Replacing bus drivers
Answer: a

56. Assembly language uses:
a) English-like keywords
b) Mnemonics instead of binary opcodes
c) Only decimal numbers

d) Only high-level functions
Answer: b

57. Which addressing mode specifies the operand directly in the instruction?
a) Register
b) Register indirect
c) Immediate
d) Indexed
Answer: c

58. In register addressing mode, the operand is:
a) In memory
b) In a CPU register
c) In IO port
d) In stack
Answer: b

59. In register-indirect addressing, the address of the operand is stored in:
a) Memory
b) A CPU register
c) Stack pointer
d) Program counter
Answer: b

60. Indexed addressing mode uses:
a) Base register + index value
b) Only base register
c) Only index value
d) Only stack pointer
Answer: a

61. A loop in assembly is typically implemented using:
a) Only `JMP`
b) Counter and conditional branch
c) Only `CALL`
d) Only interrupts
Answer: b

62. A subroutine in assembly language is usually called using:
a) `JMP`
b) `CALL` / `RET`
c) `INT`
d) `MOV`
Answer: b

63. Parameters are typically passed to a subroutine via:
a) Only data bus
b) Registers, stack, or memory
c) Only CPU cache
d) Only interrupts
Answer: b

64. In port-mapped I/O, the processor accesses I/O devices through:
a) Main memory addresses only
b) Special I/O port addresses
c) Only DMA

d) Only cache
Answer: b

65. In memory-mapped I/O, I/O devices are accessed as:
 a) Separate from memory
 b) Ordinary memory locations
 c) Only through registers
 d) Only through stack
 Answer: b
66. The main advantage of assembly language over high-level language is:
 a) Easier debugging
 b) Better control over hardware and higher efficiency
 c) Portability
 d) Simpler syntax
 Answer: b
67. The `MOV` instruction in assembly language is used for:
 a) Arithmetic operations
 b) Data transfer between registers or memory
 c) Branching
 d) I/O control
 Answer: b
68. The `JMP` instruction is used for:
 a) Data transfer
 b) Unconditional branching
 c) Arithmetic
 d) Memory allocation
 Answer: b
69. The `CMP` instruction usually:
 a) Stores data
 b) Compares two operands and updates flags
 c) Calls a subroutine
 d) Reads I/O
 Answer: b
70. The `PUSH` instruction in stack-oriented assembly:
 a) Decrements stack pointer and stores data
 b) Increments stack pointer and stores data
 c) Only reads data
 d) Only clears stack
 Answer: a
71. The `POP` instruction in stack-oriented assembly:
 a) Increments stack pointer and reads data
 b) Decrements stack pointer and reads data
 c) Only stores data
 d) Only clears stack
 Answer: a
72. A relative addressing mode is commonly used with:
 a) Unconditional jumps
 b) Branch instructions
 c) Data transfer instructions

d) Arithmetic instructions
Answer: b

73. The translator that converts assembly language into machine code is called:
a) Compiler
b) Interpreter
c) Assembler
d) Linker
Answer: c

74. In GTU-style COA syllabus, assembly-language programming is mainly studied to understand:
a) High-level GUI design
b) How instructions map to low-level CPU operations
c) Only pseudo-code writing
d) Only operating system APIs
Answer: b

75. A typical assembly program uses:
a) Only high-level constructs
b) Mnemonics, labels, and directives
c) Only decimal numbers
d) Only macros
Answer: b

76. The `ADD` instruction in assembly usually performs:
a) Bitwise AND
b) Integer addition of operands
c) Logical shift
d) Memory allocation
Answer: b

77. The `SUB` instruction typically:
a) Adds operands
b) Subtracts source from destination
c) Multiplies operands
d) Shifts bits
Answer: b

78. The `JZ` or `JE` instruction is a:
a) Unconditional jump
b) Conditional jump based on zero flag
c) Subroutine call
d) Data transfer instruction
Answer: b

79. The `IN` and `OUT` instructions are used in:
a) Only stack-based systems
b) Port-mapped I/O
c) Only memory-mapped I/O
d) Only cache
Answer: b

80. In a typical assembly loop, the loop counter is usually stored in:
a) I/O port
b) General-purpose register
c) Stack pointer only
d) Program counter

Answer: b

---

81. In a microprogrammed control unit, control signals are generated from:
    a) Hardwired logic only
    b) Control memory (microcode)
    c) Program memory
    d) Cache
    Answer: b
82. Microinstructions are stored in:
    a) Main memory
    b) Cache
    c) Control memory
    d) CPU registers
    Answer: c
83. The main advantage of microprogrammed control over hardwired control is:
    a) Higher speed
    b) Easier modification and debugging of control logic
    c) Lower power consumption
    d) Larger instruction set
    Answer: b
84. Address sequencing in microprogrammed control refers to:
    a) How the CPU accesses main memory
    b) How the next microinstruction address is determined
    c) How the stack grows
    d) How I/O devices are mapped
    Answer: b
85. A typical microinstruction includes:
    a) Only data
    b) Control signals, next-address field, and condition fields
    c) Only branch addresses
    d) Only cache tags
    Answer: b
86. In BE04000251, microprogrammed control is studied to understand:
    a) Only high-level programming
    b) Internal implementation of the control unit
    c) Only device drivers
    d) Only operating system internals
    Answer: b
87. Horizontal micro-instructions usually:
    a) Use many bits to represent many control signals
    b) Require only one bit per instruction
    c) Have compact encoding
    d) Replace ALU
    Answer: a
88. Vertical micro-instructions usually:
    a) Are wide and parallel
    b) Use encoded fields and are more compact
    c) Are slower than horizontal
    d) Only support one instruction
    Answer: b

89. The control unit must handle:
a) Only ALU operations
b) Timing, control signals, and exception handling
c) Only memory read/write
d) Only I/O devices
Answer: b
90. The concept of “microprogram” in COA is similar to:
a) High-level API
b) Firmware or low-level control-program stored in ROM
c) Operating system shell
d) Compiler
Answer: b
91. A branch microinstruction typically decides the next address based on:
a) Program counter
b) Current instruction only
c) Condition flags and control-unit state
d) Cache misses
Answer: c
92. The control-memory address register is often called:
a) MAR
b) MDR
c) CMAR (or micro-PC)
d) IR
Answer: c
93. The control-memory data register (CMDR) holds:
a) Main memory address
b) Current microinstruction
c) Program counter value
d) ALU result
Answer: b
94. Conditional branching at microinstruction level is used to:
a) Speed up main memory
b) Change microinstruction sequence based on status
c) Increase cache size
d) Replace registers
Answer: b
95. The main role of the control unit is to:
a) Only add numbers
b) Orchestrate all units using control signals
c) Only store data
d) Only display output
Answer: b
96. In microprogrammed control, the size of control memory affects:
a) Bus width
b) Flexibility and number of supported microinstructions
c) Clock speed only
d) Instruction length
Answer: b
97. The term “control store” usually refers to:
a) Main memory

b) Control memory storing microcode
c) Cache
d) I/O buffer
Answer: b

98. A horizontal micro-instruction format is:
a) More compact
b) Less parallel
c) More parallel and wide
d) Only sequential
Answer: c

99. In vertical micro-programming, the control fields are:
a) Directly wired to control lines
b) Encoded so that decoders generate control signals
c) Only for status flags
d) Only for I/O
Answer: b

100. Microprogrammed control is *not* best suited for:
a) Frequent modifications of control logic
b) Very high-speed, fixed-function pipelines
c) Educational explanation of control
d) Prototyping
Answer: b

101. In BE04000251, microprogrammed control helps students:
a) Only write assembly programs
b) Implement and trace microinstruction sequences
c) Only design I/O devices
d) Only draw diagrams
Answer: b

102. A typical microinstruction sequence for `ADD` includes:
a) Only fetch phase
b) Fetch, decode, operand-fetch, execute phases in microcode
c) Only interrupt handling
d) Only cache lines
Answer: b

103. “Next-address logic” in microprogrammed control is responsible for:
a) Fetching data from cache
b) Deciding which microinstruction comes next
c) Managing interrupts
d) Only I/O
Answer: b

104. Hardwired control is generally:
a) More flexible than microprogrammed
b) Faster but harder to modify
c) Slower and easier to modify
d) Not used in real CPUs
Answer: b

105. Microprogramming is especially useful when:
a) The instruction set is complex (CISC)
b) Instructions are very simple (RISC only)
c) Only one instruction exists

d) There is no control unit
Answer: a

106. A typical CPU contains:
a) Only ALU
b) ALU, registers, and control unit
c) Only memory
d) Only I/O
Answer: b

107. Which of the following is *not* a CPU register?
a) PC
b) IR
c) MAR
d) GPU
Answer: d

108. A general register organization allows:
a) Only one accumulator
b) Multiple general-purpose registers
c) Only stack-based operations
d) Only I/O registers
Answer: b

109. Stack-based organization uses the stack pointer primarily for:
a) Arithmetic only
b) Accessing operands and managing subroutine calls
c) Only memory-mapped I/O
d) Only interrupts
Answer: b

110. In a stack-oriented CPU, `PUSH` and `POP` operate on:
a) Main memory directly
b) The stack pointed to by SP
c) I/O ports
d) Cache tags
Answer: b

111. The addressing mode that uses the stack pointer is:
a) Immediate
b) Register
c) Stack
d) Indexed
Answer: c

112. RISC architecture emphasizes:
a) Very few registers and complex instructions
b) Simple instructions and many registers
c) Only microcoded control
d) Only one addressing mode
Answer: b

113. CISC architecture emphasizes:
a) Only load-store instructions
b) Many addressing modes and complex instructions
c) Only simple instructions
d) Only stack-based operations
Answer: b

114. In a typical CPU, the instruction pipeline is used to:
a) Increase memory size
b) Overlap execution of multiple instructions
c) Replace cache
d) Replace ALU
Answer: b
115. Pipelining can cause:
a) Only faster I/O
b) Hazards such as data, control, and structural
c) Only larger programs
d) Only simpler instructions
Answer: b
116. The stages of a classic 5-stage pipeline are:
a) Only fetch and execute
b) IF, ID, EX, MEM, WB
c) Only fetch, decode
d) Only execute and write-back
Answer: b
117. Data hazard in pipelining occurs when:
a) Instructions are too simple
b) One instruction depends on a result not yet available
c) Cache is full
d) Bus is idle
Answer: b
118. Control hazard mainly occurs due to:
a) Memory speed
b) Branch or jump instructions
c) Only I/O
d) Only cache misses
Answer: b
119. Structural hazard occurs when:
a) Two instructions need the same hardware unit at the same time
b) Only one register exists
c) Programs are small
d) Only interrupts
Answer: a
120. Typical methods to reduce pipeline hazards include:
a) Not using pipelines
b) Forwarding, branch prediction, and stall insertion
c) Only larger memory
d) Only simpler registers
Answer: b
121. Superscalar processors can:
a) Execute only one instruction per cycle
b) Issue and execute multiple instructions per cycle
c) Only run assembly programs
d) Only run CISC
Answer: b
122. The CPU performance metric “CPI” stands for:
a) Clock Per Instruction

b) Cycles Per Instruction
c) Control Per Instruction
d) Cache Per Instruction
Answer: b

123. A higher CPI generally means:
a) Faster execution
b) More clock cycles per instruction
c) Less pipeline stages
d) Simpler instructions
Answer: b

124. The main benefit of a multi-core CPU is:
a) Only larger memory
b) Running multiple threads or programs in parallel
c) Only better graphics
d) Only simpler I/O
Answer: b

125. In BE04000251, CPU organization is studied to understand:
a) Only high-level algorithms
b) Register organizations, pipelining, and RISC/CISC
c) Only operating systems
d) Only networking
Answer: b

126. The instruction register (IR) mainly holds:
a) Address of memory
b) Current instruction being decoded or executed
c) ALU result
d) Stack pointer
Answer: b

127. The status (flag) register stores:
a) Only data
b) Flags such as zero, carry, sign, overflow
c) Only addresses
d) Only cache tags
Answer: b

128. The program counter is used for:
a) Storing data
b) Holding address of next instruction
c) Only I/O
d) Only cache
Answer: b

129. Register windows are mainly used to:
a) Hold floating-point values only
b) Speed up procedure calls in some RISC designs
c) Replace cache
d) Only store I/O data
Answer: b

130. A load-store (RISC) architecture typically:
a) Allows arithmetic directly on memory operands
b) Restricts arithmetic to register operands only
c) Eliminates registers

d) Eliminates cache

Answer: b

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131. The main idea behind memory hierarchy is to:

a) Use one large memory only
b) Use faster, smaller memory near CPU and slower, larger memory away
c) Eliminate cache
d) Use only disk as main memory

Answer: b

132. Cache memory is placed between:

a) CPU and disk
b) CPU and main memory
c) Main memory and I/O
d) I/O and disk

Answer: b

133. Virtual memory allows:

a) Running programs larger than physical RAM
b) Only small programs
c) Only programs equal to RAM size
d) Eliminating page faults

Answer: a

134. Associative (content-addressable) memory is mainly used for:

a) Storing images
b) Content-addressable storage such as cache tags or TLB
c) Only main memory
d) Only I/O buffering

Answer: b

135. Typical cache mapping techniques include:

a) Direct, associative, set-associative
b) Only direct
c) Only virtual
d) Only stack

Answer: a

136. In direct-mapped cache, each block of main memory maps to:

a) Any set
b) One specific cache line
c) Only one word
d) Full cache

Answer: b

137. In fully associative cache, a block can be placed:

a) In one fixed line
b) In any cache line
c) Only in odd lines
d) Only in even lines

Answer: b

138. In set-associative cache, a block maps to:

a) One specific line
b) One of a set of lines
c) All lines

d) Only one word

Answer: b

139. A cache “hit” means:

a) Data is not in cache
b) Data is found in cache
c) Cache is empty
d) Cache is full

Answer: b

140. A cache “miss” means:

a) Data is in cache
b) Data must be fetched from slower memory
c) CPU is stopped
d) Pipeline is empty

Answer: b

141. Cache hit rate is defined as:

a) (Misses / total accesses)
b) (Hits / total accesses)
c) (Total accesses / hits)
d) (Misses / hits)

Answer: b

142. TLB stands for:

a) Top-Level Bus
b) Translation Lookaside Buffer
c) Total Line Buffer
d) Target Logic Block

Answer: b

143. TLB is used to speed up:

a) Disk I/O
b) Virtual-to-physical address translation
c) Only assembly code
d) Only cache

Answer: b

144. In virtual memory, a “page” is:

a) A small disk chunk
b) A fixed-size block of virtual memory
c) A cache line
d) An I/O port

Answer: b

145. A page fault occurs when:

a) Cache is full
b) A referenced page is not in main memory
c) Pipe is full
d) Only interrupts

Answer: b

146. The main external memory technology in COA context is:

a) Cache
b) RAM
c) Disk (HDD/SSD)
d) Registers

Answer: c

147. The term “memory hierarchy” usually includes:
a) Registers, cache, main memory, secondary memory
b) CPU only
c) Only cache and main memory
d) Only disk
Answer: a
148. The main characteristic of cache memory is:
a) Very large, very cheap
b) Very fast, small, and expensive per bit
c) Very slow
d) Only used for I/O
Answer: b
149. DRAM is used mainly for:
a) Cache
b) Main memory
c) Registers
d) TLB
Answer: b
150. SRAM is used mainly for:
a) Main memory
b) Cache
c) Disk
d) Registers only
Answer: b
151. The average memory access time (AMAT) depends on:
a) Cache hit rate and hit/miss penalties
b) Only CPU clock
c) Only disk speed
d) Only instruction length
Answer: a
152. Write-through cache policy means:
a) Writes go to cache only
b) Writes go to cache and main memory both
c) Writes go to main memory only
d) Writes are ignored
Answer: b
153. Write-back cache policy means:
a) Writes update only main memory
b) Writes update cache and mark block dirty, main memory updated on eviction
c) Writes are ignored
d) Writes go only to I/O
Answer: b
154. A replacement policy in cache is used when:
a) Cache is empty
b) Cache is full and a new block must be loaded
c) CPU halts
d) Only on reset
Answer: b

155. Common cache replacement policies are:

a) FIFO, LRU, Random
b) Only FIFO
c) Only LIFO
d) Only random

Answer: a

156. Locality of reference in memory access refers to:

a) Only random access
b) Spatial and temporal locality of program references
c) Only hardware locality
d) Only disk locality

Answer: b

157. The working set of a process in virtual memory is:

a) All pages in disk
b) The set of pages actively used during a time interval
c) Only cache lines
d) Only I/O buffers

Answer: b

158. Thrashing in virtual memory occurs when:

a) Cache is too large
b) System spends more time swapping pages than executing instructions
c) Disk is small
d) Only CPU is slow

Answer: b

159. In BE04000251, memory organization topics include:

a) Only registers
b) Cache, virtual memory, TLB, and external memory
c) Only arithmetic
d) Only networking

Answer: b

160. A memory interleaving scheme is used mainly to:

a) Increase logical address space
b) Increase effective memory bandwidth by parallel banks
c) Decrease cache
d) Only expand disk

Answer: b

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161. Booth's algorithm is used for:

a) Unsigned addition
b) Signed multiplication
c) Division only
d) Only floating-point addition

Answer: b

162. In Booth's algorithm, consecutive 1s in multiplier are:

a) Added individually
b) Handled efficiently by recoding to add/subtract and shift
c) Ignored
d) Only shifted

Answer: b

163. In floating-point representation, the exponent is usually stored in:
a) Sign-magnitude
b) 1’s complement
c) 2’s complement
d) Biased (excess-N) form
Answer: d
164. The IEEE 754 standard for single-precision floating-point uses:
a) 1 sign bit, 7 exponent bits, 24 mantissa bits
b) 1 sign bit, 8 exponent bits, 23 mantissa bits
c) 1 sign bit, 8 exponent bits, 24 mantissa bits
d) 2 sign bits, 8 exponent bits, 23 mantissa bits
Answer: b
165. Normalization in floating-point representation ensures that:
a) Exponent is zero
b) Mantissa is within a fixed range with a leading 1 (in binary)
c) All bits are zero
d) Exponent is negative
Answer: b
166. A guard bit and round bit are used in floating-point arithmetic to:
a) Control overflow
b) Improve rounding accuracy
c) Increase exponent size
d) Reduce mantissa
Answer: b
167. BCD addition requires an adjustment when:
a) Sum nibble > 9 or carry out occurs
b) Sum nibble < 9
c) No carry occurs
d) Only negative numbers
Answer: a
168. Decimal arithmetic in COA generally refers to operations on:
a) Binary integers
b) ASCII codes
c) BCD-encoded decimal digits
d) Hex numbers
Answer: c
169. Overflow in signed addition occurs when:
a) Two numbers of same sign produce result of opposite sign
b) Two numbers of different sign are added
c) Result is zero
d) Only when carry is zero
Answer: a
170. In 2’s complement, subtraction $\mathrm{A}-\mathrm{B}$$\boxed{A-B}$ is implemented as:
a) A + B
b) A + (2’s complement of B)
c) B - A only
d) A shifted right
Answer: b
171. A carry out from MSB in 2’s complement addition indicates:
a) Always overflow

b) Not necessarily overflow
c) Underflow
d) Divide by zero
Answer: b

172. Fixed-point representation is best suited for:
a) Very large dynamic range
b) Simple integer arithmetic
c) Only approximate fractions
d) Only scientific computing
Answer: b

173. Floating-point representation is best when:
a) Only integers are used
b) A very large dynamic range is needed
c) Memory is very small
d) Only addresses are stored
Answer: b

174. In COA, an arithmetic logic unit (ALU) usually supports:
a) Only integer operations
b) Integer arithmetic and logical operations
c) Only floating-point operations
d) Only decimal
Answer: b

175. A carry look-ahead adder is used to:
a) Slow down addition
b) Speed up multi-bit addition by computing carries in parallel
c) Reduce number of bits
d) Convert from binary to decimal
Answer: b

176. The term “partial product” is associated with:
a) Addition
b) Multiplication
c) Division
d) Subtraction
Answer: b

177. Restoring division in binary uses:
a) Only adds
b) Trial subtraction and possible restore
c) Only shifts
d) Only adds 1
Answer: b

178. Non-restoring division improves over restoring by:
a) Avoiding all subtractions
b) Reducing or eliminating restore steps
c) Avoiding shifts
d) Using decimal arithmetic
Answer: b

179. A floating-point underflow occurs when:
a) Exponent is too large
b) Exponent is too small to be represented
c) Mantissa is zero

d) Sign bit is 1

Answer: b

180. A floating-point overflow occurs when:
a) Exponent is too small
b) Exponent exceeds maximum representable value
c) Mantissa is normalized
d) Base changes

Answer: b

181. Rounding mode “round to nearest even” is used to:
a) Always round up
b) Reduce rounding bias in floating-point
c) Always round down
d) Ignore rounding

Answer: b

182. In BE04000251, computer arithmetic topics include:
a) Only logic operations
b) Integer, fixed-point, floating-point, and decimal arithmetic
c) Only networking
d) Only OS scheduling

Answer: b

183. Sign-magnitude representation has the drawback that:
a) It uses less bits
b) It has two representations for zero
c) It cannot represent positive numbers
d) It cannot represent negative numbers

Answer: b

184. 1’s complement representation is rarely used now because:
a) It cannot represent zero
b) It is slower than sign-magnitude
c) It also has two zeros and complicates arithmetic
d) It needs more bits

Answer: c

185. In 2’s complement, the negative of a number is obtained by:
a) Inverting bits
b) Inverting bits and adding 1
c) Shifting left
d) Shifting right

Answer: b

186. A half-adder can add:
a) Two multi-bit numbers
b) Two single bits and produce sum and carry
c) One bit only
d) Two bytes

Answer: b

187. A full-adder has how many inputs (including carry-in)?
a) 2
b) 3
c) 4
d) 1

Answer: b

188. A ripple-carry adder has the disadvantage that:
a) It is too fast
b) Carry must propagate through all stages, causing delay
c) It cannot be cascaded
d) It cannot handle BCD
Answer: b

189. In COA, the arithmetic unit design is important to:
a) Only save memory
b) Meet performance and correctness requirements of operations
c) Only handle interrupts
d) Only support OS functions
Answer: b

190. A floating-point ALU must additionally handle:
a) Normalization and rounding
b) Only integer overflow
c) Only shifts
d) Only BCD
Answer: a

191. An **I/O interface** is required to:
a) Connect CPU directly to disk
b) Match characteristics of CPU and peripheral devices
c) Replace main memory
d) Replace control unit
Answer: b

192. Asynchronous data transfer between CPU and device implies:
a) Both operate under same clock
b) Timing is not coordinated by a common clock; handshaking signals are used
c) No control signals are needed
d) Data is not acknowledged
Answer: b

193. In **programmed I/O**, data transfer is controlled by:
a) Hardware automatically
b) CPU polling status of I/O devices
c) DMA controller
d) I/O processor only
Answer: b

194. In **interrupt-driven I/O**, CPU:
a) Polls continuously
b) Is interrupted when device is ready
c) Never interacts with I/O
d) Transfers data directly to disk
Answer: b

195. In **DMA** (Direct Memory Access), data is transferred:
a) By CPU instruction by instruction
b) Directly between memory and I/O device with minimal CPU intervention
c) Only through cache
d) Only through stack
Answer: b

196. The **DMA controller** needs to know:
a) Only device speed
b) Memory address, word count, and direction of transfer
c) Instruction format
d) Pipeline depth
Answer: b

197. A **priority interrupt** system is used to:
a) Disable all interrupts
b) Decide which interrupt to service first when multiple occur
c) Sort data
d) Manage memory
Answer: b

198. A **vectored interrupt** provides:
a) Same address for all interrupts
b) Different service routine address for each interrupt source
c) No address at all
d) Only software polling
Answer: b

199. In **daisy-chain priority**:
a) All devices have equal priority
b) Priority is determined by physical order in the chain
c) Priority is random
d) CPU decides each time
Answer: b

200. An **I/O processor (IOP)** is:
a) A DMA controller with its own instruction set to handle I/O
b) A hard disk
c) A GPU
d) A stack pointer
Answer: a

201. CPU-IOP communication usually uses:
a) Shared memory and command/status words
b) No protocol
c) Only interrupts
d) Only disk
Answer: a

202. Serial communication sends data:
a) Many bits in parallel at once
b) One bit at a time over a single line
c) Only bytes at a time
d) Only words at a time
Answer: b

203. Asynchronous serial communication uses:
a) Shared clock for sender and receiver
b) Start and stop bits for each character
c) Only parity bits
d) Only DMA
Answer: b

204. Synchronous serial communication:
a) Needs no clock

b) Uses shared clock and sends blocks of data
c) Sends only one character
d) Always uses parity
Answer: b

205. Modes of transfer include:
a) Programmed I/O, Interrupt-driven I/O, DMA
b) Only Programmed I/O
c) Only DMA
d) Only Interrupts
Answer: a

206. In BE04000251, the I/O unit emphasizes:
a) Only device physics
b) Interfaces, transfer modes, interrupt/DMA, IOP, serial communication
c) Only GUI
d) Only OS drivers
Answer: b

207. A **multiprocessor** system is one that:
a) Has one CPU and many disks
b) Contains two or more CPUs sharing memory and/or I/O
c) Has many independent PCs connected by a network
d) Has only one ALU
Answer: b

208. A **shared-memory multiprocessor** means:
a) Each CPU has its own completely private memory only
b) Multiple CPUs access a common address space
c) Only disk is shared
d) Only cache is shared
Answer: b

209. A major advantage of shared-memory multiprocessors is:
a) Very simple programming model using shared variables
b) No synchronization issues
c) No need for cache
d) No scalability issues
Answer: a

210. A **message-passing multicomputer** differs from shared-memory multiprocessor in that:
a) No CPUs
b) Nodes communicate via explicit messages, not shared variables
c) No memory
d) Only one bus
Answer: b

211. An **interconnection structure** in multiprocessors refers to:
a) Type of ALU
b) How processors and memories are physically connected (bus, crossbar, network)
c) How cache is designed
d) OS structure
Answer: b

212. A **bus-based multiprocessor** is limited mainly by:
a) Complexity of bus protocol

b) Bus bandwidth as more processors are added
c) Lack of cache
d) Lack of memory
Answer: b

213. A **crossbar switch** interconnection offers:
a) Simple, single shared path
b) Dedicated path between any processor–memory pair
c) No parallelism
d) Only serial links
Answer: b

214. **Inter-processor arbitration** is needed to:
a) Allocate CPU time to interrupts
b) Decide which processor gets access to a shared bus or resource
c) Decide instruction format
d) Choose cache size
Answer: b

215. Inter-processor communication in shared memory usually uses:
a) Shared variables in memory, plus synchronization primitives
b) Only I/O ports
c) Only network packets
d) Only DMA
Answer: a

216. Synchronization between processors often relies on:
a) Delays only
b) Locks, semaphores, barriers, atomic operations
c) Only interrupts
d) Only cache
Answer: b

217. **Cache coherence** refers to:
a) All caches having the same size
b) Keeping multiple cached copies of same memory location consistent
c) Having no cache
d) Only write-through policy
Answer: b

218. A simple coherence problem arises when:
a) A variable is never used
b) One processor updates data in its cache; others still see the old value
c) Only one CPU exists
d) No cache exists
Answer: b

219. A common hardware cache-coherence method is:
a) Write-back only
b) Snooping protocols on a shared bus
c) Only FIFO cache
d) Only TLB
Answer: b

220. **MESI** protocol is used for:
a) Disk control
b) Cache coherence (Modified, Exclusive, Shared, Invalid states)
c) TLB replacement

d) Virtual memory
Answer: b

221. A **shared-memory multiprocessor** can be classified in Flynn's taxonomy as:
a) SISD
b) SIMD
c) MIMD
d) MISD
Answer: c

222. In BE04000251, the Multiprocessors unit focuses on:
a) Only single-core CPUs
b) Characteristics, interconnection, arbitration, communication, and coherence in multiprocessors
c) Only scalar arithmetic
d) Only serial ports
Answer: b